Gaurav Kumar Gautam

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RESEARCH PROFILE

I am a Geomatics Engineering researcher specializing in multisource remote sensing, InSAR deformation monitoring, and machine learning for resource exploration, geohazard assessment, and mining-environment monitoring. My work integrates SAR (PSInSAR/SBAS time-series analysis), hyperspectral and optical imagery, thermal and nighttime-light data, spatial statistics, deep learning segmentation and restoration targeting, and generative diffusion models for data-scarce settings. I seek doctoral research that advances Earth-observation and AI methods for mineral and energy resource exploration and surface-deformation hazard monitoring, with a continuing interest in the socio-environmental and community-level dimensions of mining.

EDUCATION

Aug 2024 – May 2026

M.Tech, Geomatic Engineering

Indian Institute of Technology (ISM), Dhanbad — CGPA 8.48/10.00

Thesis: Multi-Temporal and Spatial Autocorrelation-Based Assessment of Mining-Induced Ecological Degradation Using a Restoration Priority Index: A Case Study of the Jharia Coalfield, India.

Supervisor: Prof. Dheeraj Kumar, Professor (HAG), Department of Mining Engineering.

Developed the first integrated decadal restoration prioritization framework for the Jharia Coalfield, a ~370 km² landscape affected by a century of mining and underground coal-seam fires. We fused six degradation indicators across a ten-year satellite archive, introducing two novel indices: *Vol-NDVI* (distinguishing structural vegetation from ephemeral grasses) and *Surface Alteration Stress* (quantifying anthropogenic mining pressure). The resulting spatially validated Restoration Priority Index (Moran's I = 0.8831, ROC–AUC = 0.8355) identifies, at a 10-meter resolution, where restoration is most urgent and why.

Sep 2019 – Sep 2023

B.Tech, Civil Engineering

Bihar Engineering University, Patna — CGPA 7.99/10.00

Final Year Project: Modifying the GRIHA Rating System for Existing Buildings: A Case Study on Government Engineering College, Arwal — *awarded Best Final Year Project, 2022–23.*

The program established the basics of structural analysis, hydraulics, geotechnical and environmental engineering, and construction technology. In the final year, the project modified GRIHA, India’s national green-building standard for new buildings, for an existing institutional campus. It evaluates energy, water, waste, and site ecology to identify practical improvements. This early focus on sustainability in the built environment helped move toward the development of environmental geomatics.

GRANTS AND AWARDS

2024–2026

Teaching Assistantship, IIT (ISM), Dhanbad. A competitive award based on merit obtained through India's national postgraduate entrance exam. It offers tuition support and a monthly research stipend and involves undergraduate teaching duties in the Department of Mining Engineering, and

2022–2023 **Best Final Year Project Award**, Bihar Engineering University, Patna, India. Awarded for *Modifying the GRIHA Rating System for Existing Buildings.*

2019–2023 **Chief Minister's Merit-cum-Support Scholarship for Higher Education**, Government of Bihar, India. A competitive state scholarship awarded on academic merit, supporting four years of undergraduate study.

RESEARCH EXPERIENCE AND PROJECTS

Apr 2026 – Present

Researcher — GCT-TabDDPM: Geo-Conditional Diffusion Model for Landslide Augmentation

IIT (ISM) Dhanbad

Objective: Addressing chronic data scarcity in geohazard modeling by augmenting sparse landslide inventories in the Sikkim Himalaya.
Methodology: Development of a Geo-Conditional Transformer coupled with a Tabular Denoising Diffusion Probabilistic Model under ten domain-specific geomechanical constraints.
Impact: Statistically faithful synthetic samples were generated to overcome class imbalance and spatial bias. Manuscript in preparation for the *International Journal of Disaster Risk Reduction*.

Mar 2026 – present

Researcher — Multi-Temporal InSAR Landslide Monitoring, Gangtok, Sikkim

Dept of Mining Engineering, IIT (ISM) Dhanbad

We conducted a multi-temporal InSAR characterization of ground-surface deformation across the Gangtok region, addressing the lack of pixel-scale monitoring in vegetated Himalayan urban centers. A full year of Sentinel-1A ascending and descending acquisitions (55 scenes) was processed through the ISCE2/MintPy pipeline, applying both SBAS and Persistent Scatterer InSAR to generate 200 interferometric pairs and approximately 10 million coherent pixels. Joint inversion decomposed the line-of-sight displacement into horizontal and vertical components, resolving the 3D kinematics of active slow-moving landslides. Time-series analysis showed a three- to five-fold increase in speed during the June–September monsoon, offering quantitative proof that pore pressure from water affects slope instability. Manuscript ready for submission.

Jan 2025 – Apr 2026

Graduate Researcher (M.Tech Thesis) — Jharia Coalfield Restoration Prioritization

Dept. of Mining Engineering, IIT (ISM) Dhanbad · Supervisor: Prof. Dheeraj Kumar

Developed the first integrated decadal restoration prioritization framework for the Jharia Coalfield, a ~370 km² landscape affected by a century of mining and persistent coal-seam fires. I fused six degradation indicators from a ten-year archive (integrated via Theil–Sen slope estimation with Mann–Kendall significance) of Sentinel-1/2, Landsat-8/9, and VIIRS data, introducing two new indices (a SAR–optical fusion index and a Surface Alteration Stress index), and weighted them using Fuzzy AHP with multicollinearity control. Segmentation using the GeoAI model U-Net was employed to identify the target area for restoration**.** The resulting Restoration Priority Index was validated across four independent dimensions (Moran's I = 0.8831; ROC–AUC = 0.8355; Monte Carlo CV = 2.56%; 77.48% coal-fire capture), providing reclamation planners with a 10-meter map of where restoration is most urgent and why. {Manuscript submitted to *Science of Remote Sensing* (Elsevier, 2025); code and data openly archived on GitHub.}

Summer 2025

Independent Researcher — Hyperspectral Mineral Mapping, Bailadila Iron Ore Mine

Self-directed work during M.Tech recess

Built an EO-1 Hyperion processing workflow in ENVI (FLAASH correction, PCA/MNF reduction, Spectral Angle Mapper) to map hematite, pyrite, and quartz-associated minerals, extending the methodological breadth in spaceborne spectroscopy.

Aug 2024 – Mar 2025

Graduate Researcher — Landslide Susceptibility Modeling, Sikkim Himalaya

IIT (ISM) Dhanbad

Objective: To conduct a three-phase investigation of landslide susceptibility in the tectonically active Sikkim Himalaya.

Methodology: In Phase 1, GIS-based zonation was developed using parameterized AHP, introducing 14 predictor layers and 672 landslide points. Phase 2 benchmarked AHP, Frequency Ratio, Logistic Regression, and hybrid models. Phase 3 evaluated ensemble machine learning algorithms (XGBoost and WSRF) using 5-fold spatial cross-validation.

Impact: XGBoost achieved the highest discrimination (AUC = 0.895). Partial dependence analysis identified a critical slope-instability transition between 25° and 30°, which aligns with geomorphological friction theory.

Mar 2025 – Aug 2025

Data Science Intern — TEXMiN, IIT (ISM) Dhanbad

Technology Innovation Hub for Excellence in Mining

Contributed to the predictive-monitoring program of TEXMiN, a national Mining Technology Innovation Hub, building Python machine-learning pipelines that extract environmental degradation metrics from high-resolution satellite imagery in support of data-driven mine management.

Nov 2024 – Mar 2025

Contributing Researcher — Groundwater Potential Zonation, Purulia, West Bengal

Collaborative manuscript project

We benchmarked AHP, Frequency Ratio, Logistic Regression, and Random Forest against 66 verified well observations across the hard-rock aquifers of the Chota Nagpur Plateau, finding that data-driven models (AUC = 0.75) outperformed expert weighting (AUC = 0.66). Methodology relevant to India's National Aquifer Mapping Programme (NAQUIM).

Sep 2024 – Feb 2025

Contributing Researcher — Integrated Coastal Flood Risk Assessment, Sagar Island

Collaborative book chapter, Springer (In Press, 2025)

Co-authored a flood-risk framework for Sagar Island in the Indian Sundarbans, fusing remote-sensing exposure metrics with census-based socioeconomic vulnerability indicators to map priority intervention zones, extending my work into the socio-environmental dimensions of climate hazards.

Oct 2021 – Dec 2021

Civil Engineering Intern — Technoculture Building Center Pvt. Ltd

Contributed to structural audits and green-building feasibility studies for residential developments, assisting in field surveys, energy-consumption assessment, and site-ecology data consolidation.

PUBLICATIONS

Peer-Reviewed Journal Article

Das, M., **Gautam, G. K.**, Jain, S., Bhat, M. F., Mankar, A. K., & Koner, R. (2026). A comparative analysis of AHP, FR, AHP–FR and LR models for landslide susceptibility mapping in Sikkim Himalaya, India. *Earth Surface Processes and Landforms,* 51(2), e70257.

Under Review

Gautam, G. K., Das, M., & Kumar, D. (2026). Multi-temporal and spatial-autocorrelation-based assessment of mining-induced ecological degradation using a Restoration Priority Index: A case study of the Jharia Coalfield, India. *Science of Remote Sensing* (Elsevier). *[first author]*

Das, M., **Gautam, G. K.**, & Koner, R. (2025). Benchmarking classical, ensemble, and subspace-based algorithms for landslide susceptibility mapping in the Sikkim Himalaya. *Earth Surface Processes and Landforms* (Wiley).

Das, M., **Gautam, G. K.**, & Koner, R. (2025). Modeling landslide susceptibility in the rugged terrain of Sikkim, India. *Geological Journal* (Wiley).

Das, M., Dutta, B., Mal, S., Laskar, S., Das, B., & **Gautam, G. K.** (2025). Groundwater potential zonation mapping using deterministic, statistical, and machine-learning approaches: A comparative study of Purulia District, West Bengal, India. *Arabian Journal of Geosciences* (Springer).

Book Chapter (In Press)

Patra, S., **Gautam, G. K.**, Das, M., Samanta, D., Jain, S., Bhat, M. F., & Gayen, A. (2025). Integrated coastal flood risk assessment using a multi-criteria GIS approach: A case study of Sagar Island, India. In *Sustainable Solutions.* Springer (In Press).

In Preparation

Gautam, G. K., et al. (2026). GCT-TabDDPM: A geo-conditional diffusion model for landslide inventory augmentation in the Sikkim Himalaya. Target journals: *International Journal of Disaster Risk Reduction; Computers & Geosciences.*

Das, M**., Gautam, G**. K., et al. (2026). Multi-temporal InSAR analysis of landslide deformation in Gangtok, Sikkim, using Sentinel-1A: a PSInSAR approach with ascending and descending orbit integration. [journal — to confirm]

RESEARCH SKILLS

Remote sensing & satellite data

Multi-source acquisition and preprocessing across optical, SAR, thermal, and nighttime-light sensors (Landsat-8/9, Sentinel-1/2, EO-1 Hyperion, and VIIRS); atmospheric correction, dimensionality reduction (PCA, MNF), spectral-angle classification, and cloud-based processing in the Google Earth Engine Code Editor. *Software:* ENVI, ERDAS Imagine, SNAP, and Google Earth Engine.

Geospatial modelling & GIS

Spatial overlay, raster algebra, terrain modeling, hydrological analysis, and cartographic production are used for environmental and mining applications. *Software:* ArcGIS, QGIS, CAMPAD.

Programming & scientific computing

Python geospatial and scientific stack (GeoPandas, GeoAI, Rasterio, Shapely, NumPy, SciPy, pandas, Matplotlib, OpenCV), and JavaScript for Google Earth Engine workflows.

Machine learning & deep learning

Gradient-boosted ensembles (XGBoost), random forests, weighted subspace random forests, logistic regression, and generative diffusion models (Tabular DDPM with geo-conditional constraints). *Frameworks:* PyTorch, TensorFlow, and scikit-learn.

Statistical & spatial methods

Fuzzy AHP with VIF multicollinearity control; Theil–Sen slope estimation and Mann–Kendall testing; Moran's I and Getis–Ord Gi* autocorrelation; Monte Carlo sensitivity analysis; ROC–AUC and spatial cross-validation.

ADDITIONAL SKILLS

Data management	Spatial database management and querying (SQL, PostGIS), dashboards and reporting in Power BI, and tabular analysis in Microsoft Excel.
Languages	English (full professional proficiency) and Hindi (native).

INTERESTS AND ACTIVITIES

Chess	Intra-college (IIT (ISM) Dhanbad) General Championship winner, chess team captain, and named best player at a Bihar Engineering University inter-college tournament. Member, Black Knight Chess Club, IIT (ISM), Dhanbad.
Sports leadership	Sports General Secretary, Bihar Engineering University (2022–23); Coordinator, Badminton Club, IIT (ISM) Dhanbad. Active badminton and tennis players.
Community engagement	Volunteer teacher with KARTAVYA (kartavya.org), a registered NGO providing educational outreach at IIT (ISM) Dhanbad. Teach Physics, Chemistry, and Mathematics to students in grades 9–12 from underprivileged backgrounds on weekends (2025–present).
Trekking & fieldwork	Completed the Annapurna Base Camp and Mardi Himal Himalayan treks; field visits to active coal-mining areas and landslide-affected terrain in Sikkim during research.
Other	Sports analysis, particularly football and cricket.

REFEREES

M.Tech Thesis Supervisor

Prof. Dheeraj Kumar
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Thesis Co-Supervisor

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B.Tech Project Supervisor

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